TCS — Aptitude Test 2

Questions 26–50 of 100 · 25 questions

Question 26. Find the missing number in the series 70, 54, 45, 41, ___ .

Solution:
70 – 54 = 16
54 – 45 = 9
45 – 41 = 4
The difference between the successive numbers
are
square
values
in
the
descending order.
The last square value is 4 which is the square of 2.
The next square value will be square of 1 i.e. 1.
41 – 1 = 40
Missing number = 40

Question 27. Mr. A took up the sum of natural numbers, by mistaken he add a number twice and find the sum to be 1000.Find the repeated number.

Solution:
The formula to find out sum of n natural numbers = n(n+1)/2
Substitute values for n, so that the sum will be almost equal to 1000 but less than 1000.
If n = 44, sum = 990.
To this 990 if 10 is added, the sum occurred due to error will be 1000.
The number which is added twice = 10.

T C S

Question 28. What is the 50th term of term of the series 1,3,6,10,15,21?

Solution:
First number in the series is sum of first 1 natural number.
Second number in the series is sum of first 2 natural numbers.
Third number in the series is sum
of
first 3 natural numbers.
So, 50th term of the series will be sum of

first 50 natural numbers.
50th term = 50(50+1)/2
= 1275.

Question 29. What are the last three numbers of the series 11234783131___?

Solution:
23 – 11 = 12
47 – 23 = 24
83 – 47 = 36
131 – 83 = 48
X – 131 = 60
X = 191. Last three digits = 191.

Question 30. If 212=25; 213=36; 214=47; 215=58; 216=69; 218=??

Solution:
212 = 25. Unit digit of 212 becomes tens digit, to that sum of the digits of 212 is added.
218 = 80 + 11 = 91

Question 31. Average age of students of an adult school is 40 years. 120 new students whose average age is 32 years joined the school. As a result the average age is decreased by 4 years. Find the number of students of the school after joining of the new students.

Question 32. Mean of three numbers is 10 more than the least of the numbers and 15 less than greatest of the three numbers. The median is five. Find the sum of the three numbers.

Solution:
(X + Y + Z)/3 = Mean (M)
Y = 5; M = X + 10■X = M – 10
M = Z – 15■Z = M + 15
(M – 10 + 5 + M + 15)/3 = M
■M = 10 ■X = 0; Z = 25
X + Y + Z = 0 + 5 + 25 = 30

[n(n+1)/2] – x -(x + 1)

= 105

4

n - 2

83
76

79

3
4
:

Ratio between number of students in X and Y = 3 : 4
Ratio between number of students in class Y and Z

76
85

81

4
5
:

Ratio between number of students in Y and Z = 4 : 5
Ratio between number of students in X, Y and Z = 3 : 4 : 5
Assume that the number of students in class X, Y
and Z are 3, 4 and 5
respectively.
Overall average = overall total/overall numbers.
Total marks by students in X
= 3 x 83 = 249
Total marks by students in Y
= 4 x 76 = 304
Total marks by students in X
= 5 x 85 = 425
Overall total = 249+304+425 = 978
Overall numbers = 3 + 4 + 5 = 12
Overall average = 978/12 = 81.33

Question 33. If two consecutive terms are removed from a series 1,2.....n ,then the average of remaining terms is 26(1/4). What is the no of terms?

Solution:
26(1/4) = 105/4

Let the numbers removed be x and x+1

Solution:
Let the actual number of students before the entry of 120 students be X.
Sum of the ages of original number of students = 40X.
Average age of old students and new students = 36.
Sum of ages of new 120 students = 120 x 32 = 3840.
From the above data, we get:

n – 2 will be a multiple of 4
If n - 2 = 4; n = 6. Does not satisfy the equation.
If n - 2 = 8; n = 10. Does not satisfy.
If n – 2 = 12; n = 14. Does not satisfy.
If n – 2 = 16; n = 18. Does not satisfy.
Going on like this, when
n – 2 = 12 x 4 = 48; n = 50
50(50+1)/2 = 1275 and
105 x 12 = 1260.
1275 – 1260 = 15.
Sum of two consecutive integers = 15.
The two numbers which are removed = 7, 8
The number of terms in the series = 50.

T C S

40X + 3840

X + 120
= 36

By solving the above equation, we get
X = 120
Alternate method:
Using allegation rule.

40
32

Question 34. Three math classes: X, Y, and Z, take an algebra test. The average score in class X is 83. The average score in class Y is 76. The average score in class Z is 85. The average score of all students in classes X and Y together is 79. The average score of all students in classes Y and Z together is 81. What is the average for all the classes?

36

4
4

The
ratio
between
the
number
of
students in each category = 4 : 4 = 1 : 1.
Therefore the number of old students
= 120

Solution:
Using allegation rule, find the ratio
between number of students in each
class.
Ratio between number of students in
class X and class Y.

Question 35. A cask contains 12 gallons of mixture of wine and water in the ratio 3:1. How much of the mixture must be drawn off and water substituted so the wine and water become half and half.

Solution:
The quantity of wine in the mixture
= 12 x (3/4) = 9 liters
The quantity of water in the mixture
= 12 – 9 = 3 liters
The result should have half and half of
wine and water.
Therefore, there must be 6 liters of wine
and 6 liters of water.
Out of 9 liters of wine, 3 liters must be
replaced with water.
The
fraction
of
the
solution
to
be
replaced by water = 3/9 = 1/3.

Question 36. If a person mixes Rs. 50/kg rice and Rs. 60/kg rice and sold them at Rs. 70/kg and earned a profit of 20%, then in what ratio should he mix the two varieties?

Solution:
Selling price of the mixture = Rs. 70
Profit % = 20
Cost Price of the mixture:
SP = 1.2 CP
CP = 70/1.2 = Rs. 58.33
Find the ratio using allegation rule.

T C S

50
60

Question 37. The shopkeeper charged Rs. 12 for a bunch of chocolate. But I bargained to shopkeeper and got two extra ones, and that made them cost one rupee for dozen less then how many chocolates I received in 12 rupees. What is the first asking price?

Solution:
Actual price of 1 chocolate = 12/x
Discount price of 1 chocolate = 12/x+2
Actual price of 12 chocolates = 144/x
Discount price of 12 chocs = 144/x+2
Given,
(144/x) – (144/x+2) = 1
By solving this, we get x = 16.
Actual price of the chocolate
= 12/16
= Rs. 0.75

Question 38. If apples are bought at the rate of 30 for Rs.100. How many apples must be sold for Rs.100 so as to gain 20%?

Solution:
Cost price of one apple = 100/30 = 3.33
X be the number of apples sold at 20%

Question 39. A batsman scored 110 runs which includes 3 boundaries and 8 sixes. What percent of his total score did he make by running between the wickets?

Solution:
Runs scored through boundaries and sixes = 3x4 + 8x6 = 12 + 48 = 60.
Runs scored by running between the wickets = 110 – 60 = 50
Percentage of runs scored by running between the wickets,
= (50/110) x 100 = 45.45%

Solution:
Actual price of the wheat at faulty weight is
= 0.9 x 3
= Rs. 2.70
Discount price of the wheat (@ 10%)

T C S

= 3 – (10/100) x 3
= Rs. 2.70
The actual price and discount price are both same and yet Eesha did not get any profit or loss.
Therefore, Eesha sold the wheat at cost price.
Cost price of the wheat is:
Rs. 2.70 for 0.9 kg
= Rs. 3 per kg.

Question 40. In June, a baseball team that played 60 games had won 30% of its game played. After a phenomenal winning streak, this team raised its average to 50%. How many games must the team have won in a row to attain this percentage?

Solution:
Number of matches won in the first 60 matches = (30/100) x 60 = 18
Let the number matches played after the 60th match be N.
Given, after playing N matches, the winning percentage becomes 50.
So,
[(18+N)/(60+N)] = 50/100
1800 + 100N = 3000 + 50N
50N = 1200
N = 24.

Question 41. In an examination 80% of student passed in mathematics. 55% in English. 29% failed in both subject. The pass percent in both subjects is?

Solution:
Percentage of students passed in at least one subject = 100 – 29 = 71%
Students passed in both
= 80+55-71 = 64%

Question 42. Eesha has a wheat business. She purchases wheat from a local wholesaler at a particular cost per kilogram. The

price of the wheat at her stores is Rs. 3 per kg. Her faulty spring balance reads 0.9 kg for 1 kg. Also in the festival season, she gives a 10% discount on the wheat. She found that she made neither a profit nor a loss in the festival season. At what price did Eesha purchase the wheat from the wholesaler ?

profit.
Cost price of X apples = 100/1.2 = 83.33
Number of apples which cost 83.33
= 83.33/3.33 = 25

Question 43. The ages of the two friends were in the ratio of 2:3. If the sum of their ages is 55.Then after how many years their ratio will become 4:5?

Solution:
Since the present age of A and B are in the ratio 2 : 3 we can assume that the actual present age od A and B as,
2x and 3x
2x + 3x = 55
x = 11
Age of A = 22
Age of B = 33
Let n be the number of years after which their ages will be in the ratio 4 : 5.
Therefore,
(22 + n)/(33 + n) = 4/5
5(22 + n) = 4(33 + n)
110 + 5n = 132 + 4n
5n – 4n = 132 – 110
n = 22

Question 44. When Usha was thrice as old as Nisha, her sister Asha was 25. When Nisha was half as old as Asha, then sister Usha was 34. Their present ages add to 100. How old is Usha?

Solution:
When Nisha's age is x, Usha's age is 3x and Asha's age is 25
x, 3x, 25
When Asha is 2y, Nisha is y and Usha is 34.
Subtract the ages of each person in both the cases:
x – y = 3x – 34 = 25 – 2y
2x + y = 34
3x + 2y = 59
By solving the two equations, we get

x = 9 and y = 16
When Usha is 34, Nisha is 16 and Asha is 32.
34 + 16 + 32 = 82
82 is 18 short of 100.
Add 6 years to each person.
The present age of Usha = 34 + 6 = 40.

Question 45. Curious Elva asked her father what he would gift for her nineteenth birthday. Father replied that it would depend on the day of the week and be one of SUNglasses, MONeybag, ..., FRIedcake, and SATchel. Please help Elva find the day of the week on 08-Jan-2029.

Solution:
Remember this day of the date always.
Jan 1, 2001 is Monday.
From Jan 1, 2001 to Dec 31, 2028, there are 28 years.
In these 28 years, there are 21 normal years and 7 leap years.
Number of odd days for 21 normal years = 21 x 1 = 21
Number of odd days for 7 leap years = 7 x 2 = 14
Total odd days + 8 days in 2029 = 43
Reminder of 43/7 = 1.
Since looking from past to future, add one day to Monday. Answer: Tuesday

Question 46. If there are only 4 Sundays and 4 Thursdays in the month of January then what day will be on 31st Jan?

Solution:
If a month has 31 days, there will be exactly 3 days out of Sunday to Saturday which will occur five times.
Between Thursday and Sunday, there are only two days(Friday and Saturday).
But
between

Sunday
and
Thursday,
there are three days(Monday, Tuesday
and Wednesday).
There are 3 Mondays, 3 Tuesdays and 3
Wednesdays in that month.
The last day of the month is Wednesday.

T C S

Question 47. The famous church in the city of Kumbakonam has a big clock tower and is said to be over 300 years old. Every Monday 10.00 A M the clock is set by Antony, doing service in the church. The clock loses 6 mins every hour. What will be the actual time when the faulty clock shows 3 P.M on Friday?

Solution:
For every 60 minutes of the actual time,
the faulty time is 54 minutes.
The faulty time from 10:00AM Monday
to 3:00PM Friday is:
(14 + 24 + 24 + 24 + 15) = 101 hours.
The ratio between faulty time and actual
time is 54 : 60.
therefore,

54

60
=
101

X

Where X is the actual time when faulty
time is 3:00PM
X = 112.22 = 112 hr 13 min 20 sec
Actual time is 02:22:20AM Saturday.

Question 48. George does 3/5 of a piece of work in 9 days. He then calls Paul and they finish the work in 4 days. How long would Paul take to do the work by himself?

Solution:
Time taken by George to complete 1/5th

Question 49. The five tyres of a car (four road tyres and one spare) were used equally in a journey of 40,000 km. The number of km of use of each tyre was?

Solution:
The number of tyres on the road at any time is 4.
The total distance of each of the four tyres facing the road = 40000.
Total distance by the tyres facing the road = 4 x 40000 = 160000.
The average distance for each of the 5 tyres = 160000/5 = 32000

Question 50. A certain quantity of rice is spent daily for 30 students in a hostel. One day some students were absent as a result, the quantity of rice has been spent in the ratio of 6 : 5. How many students were present on that day?

Solution:
Let N be the number of hours required to fill the tank.
The first pipe is opened for N hours.
Second pipe is opened for N – 1 hours.
Third pipe is opened for N – 2 fours.

Solution:
The rice per student is considered constant.
Ratio of rice used is directly proportional to the ratio of number of students available.

6

=
30

N = - 44, we are considering that the tank will never get filled, but a which is already full will get emptied in a time of 44 minutes.

X

5

X = 25.
Number of students present = 25

15
=
1

N - 2

5
–
N
10
+

N - 1

30
=
1
6N - 12

30
–
3N

2N - 2

30
+